Speed and velocity

Answers

Calculating average speed

1. The average speed (v) for a journey is calculated using the formula:

$$v = \frac{d}{t}$$

where d = distance travelled and t = time taken.

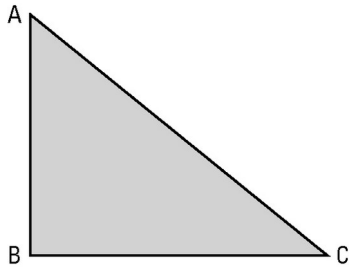
Complete the following table. Make sure you include the appropriate units for each quantity.

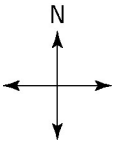
Distance travelled	Time taken	Average speed
100 m	20 s	$\frac{100}{20}$ = 5 m/s
3 km	$\frac{3}{60}$ = 0.05 h	60 km/h
20 × 15 = 300 m	15 s	20 m/s
2 km	$\frac{2000}{25}$ = 80 s	25 m/s

Average speed and average velocity

2. Two vehicles (X and Y) travel from A to C along different paths as shown in the diagram (not drawn to scale). X travels along the path ABC and vehicle Y travels along AC. The triangle ABC is a right-angled triangle. A is north of B.

AB = 600 m
BC = 800 m





Both vehicles take 70 s for the trip.
Perform the necessary calculations to complete the table.

Average speed of X	Average velocity of X	Average speed of Y	Average velocity of Y
d = 800 + 600 = 1400 m t = 70 s $v = \frac{d}{t} = \frac{1400}{70}$ = 20 m/s	$v = \frac{\text{displacement}}{\text{time}}$ $= \frac{1000}{70}$ = 14.3 m/s Direction = 37 °S of E (draw a scale diagram and measure the angle)	d(AC) = 1000 m (this is a 3:4:5 right-angled triangle) t = 70 s $v = \frac{d}{t} = \frac{1000}{70}$ = 14.3 m/s	$v = \frac{\text{displacement}}{\text{time}}$ $= \frac{1000}{70}$ = 14.3 m/s Direction = 37 °S of E (draw a scale diagram and measure the angle)

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